Zhang 2019 Follicular Lymphoma — analysis summary

RUN 1 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(θ=10) integrates the clonal FL batch: same-sample neighbour fraction 0.975 → 0.453, iLISI median 1.00 → 2.26 of 4. One connected component (0 islands). θ=2 leaves a per-patient archipelago; judged on the regenerated embedding + per-neighbourhood iLISI, not a scalar: mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

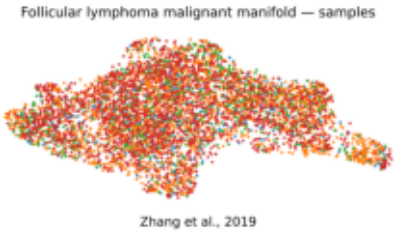
Non-branching case: a single (Gast/pSom)→Crest developmental gradient. Gast & pSom overlap at the origin (joint enrichment 1.00, smoothed r=−0.16), both oppose Crest (Gast↔Crest r=−0.55, pSom↔Crest r=−0.58). Along pseudotime Gast falls (ρ −0.48), Crest rises to a terminal tip (decile pt=0.58). Joint pSom&Crest enrichment 0.23 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — hub cross-repression → NA

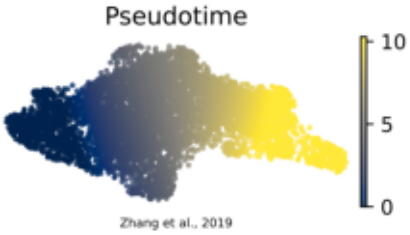
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks not produced, per spec).

STEP FIGURES

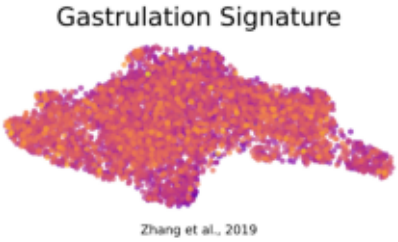
Manifold — samples



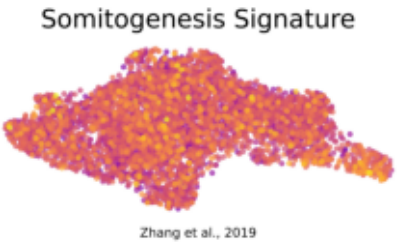
Pseudotime



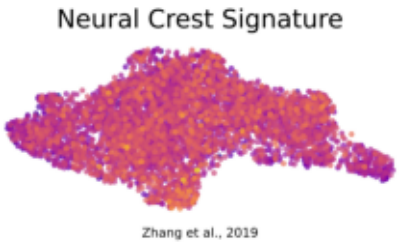
Gastrulation



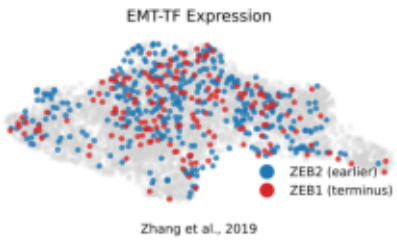
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Follicular lymphoma malignant B-cell compartment (Zhang et al., 2019), 10x — 5,645 malignant cells over 4 samples / 2 patients (primary + progressed/transformed); single low-cycling body (~14% cycling).

Verdict.
Pattern reproduces in the non-branching form (Linear-Gast): one continuous gastrulation → neural-crest axis, Crest terminal, Gast globally strongest. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB2 & ZEB1 (both terminus-loaded, ZEB2 earlier).

Parameter choices

Sample threshold	≥100 (floor; largest gap ratio 2.74, no dominant cliff) → 4/4 samples, 5,645 cells	**Overdispersed genes**	636 (scFates find_overdispersed)
Integration	Harmony on sample, θ=10 (mixing improves monotonically; clonal FL patient batch)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 0 islands removed
Geometry	Non-branching scFates curve N=20 λ=0.3 μ=100; 2 tips/0 forks, SA 2.96; pt vs cycling ρ=+0.29	**Root**	tip 2 = Gast/pSom-high, Crest-low origin
Module means (raw)	Gast +0.012 / pSom +0.006 / Crest +0.009 (Gast strongest → Linear-Gast)	**EMT-TFs shown**	ZEB2 & ZEB1 (>3% detected; both terminus-loaded, ZEB2 earlier; no root-loaded TF)